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DeepFracture: A Fine Tuned Transfer Learning based Model for Bone Fracture Classification

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***Abstract*—This work introduces DeepFracture, a transfer learning aided model which is used to classify ten different kinds of bone fractures using pre-trained architecture models VGG16, MobileNetV2, and InceptionV3. Downloaded from Kaggle, the dataset was resized and augmented as preprocessing to mitigate the issue of class imbalance. Techniques of Fine-tuning the model with respect to its architecture included: freezing certain layers of the model and learning rate up to 0.001 was reduced to 0.0001. At the end of the training MobileNetV2 was able to give the highest accuracy of 97.40%. These results prove that transfer learning can be used effectively in classifying medical images as it guarantees accurate fracture detection in a short time. This is beneficial to the healthcare practitioners as it minimizes the risks of misdiagnosis and enhances the effectiveness of treatment. Future studies might be centered on increasing the data set and applying these models in the clinical settings in real time to assist in making medical decisions.**

Keywords– Bone Fracture Classification, Transfer Learning Medical Image Processing, MobileNetV2, Deep Learning in Healthcare.

I. INTRODUCTION

AI, and deep learning more specifically, has had a lot of success so far in the area of healthcare, particularly in medical imaging and bone fracture detection, classification. Many advanced imaging techniques such as CT, MRI, and X Ray pictures require an intensive image processing and classification of normal vs abnormal patterns in order to arrive at a conclusion regarding the health status of a patient. Structures in images are easily understood by deep CNN architectures, hence able to perform deep feature extraction from medical images. These models can handle complex and voluminous data sets with medical images (X Rays) with high categorization accuracy in the identification and classification of fractures. With the automatic identification and classification of fractures using AI, the doctors are able to receive the analysis while treating the patient, thus enhancing the healing process. Another great improvement to the deep learning model in medical image classification tasks is transfer learning, which allows already learned models' usage in new specific medical tasks, without expensive data and computers.

In present times, all advancements in medical imaging techniques thanks to AI can be acknowledged. However, a few challenges still exist particularly when it comes to the quality of the data. Most medical data especially X-ray and other imaging techniques are subject to privacy issues which make it hard to come by large, labeled data to aid in the training of deep learning models. In addition, medical diagnosis images usually are Annotated by specialists, which is an expensive and time-consuming process. This limitation affects the training of accurate and generalizable models. Datasets with imbalanced classes, for example where some classes of fractures are represented while others such as pediatric fractures are represented can also lead to biased predictions from the models. These problems require data augmentation, preprocessing, and model selection in order to achieve robustness and accuracy in fracture classification tasks.

In this research, we also showcase the advancement made in the field of AI-based fracture classification by applying transfer learning models, namely VGG16, MobileNetV2, and InceptionV3 to a bone fracture dataset obtained from Kaggle. These models were optimized to identify ten different types of fractures, such as an avulsion, comminuted, and spiral fractures. The main purpose of this work is to present transfer learning, preprocessing and augmentation techniques as a viable solution to classification problems caused by insufficient data and still achieve high classification accuracy. MobileNetV2 on the other hand proved to be the best model since it achieved a remarkable 97.40% accuracy as compared to the other models experimented on underscoring its appropriateness for the task at hand.

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The structure of this paper is as follows: The introduction gives a background on the incorporation of AI and deep learning in the classification of bone fractures, together with the challenges that exist in this field. Next is the literature review part which looks into the past as well as modern methods of AI based fracture detection. The methodology describes the approach we have used in this study, outlining the dataset, preprocessing, and model selection. The result analysis section presents the results obtained from the various models and their interpretations, and the last part of the article presents the general conclusions and the prospects for the improvement of AI-based bone fracture classification systems.

II. RELATED WORK

Various works have been done on bone fracture classification whether authors have used methodologies like machine learning and dep learning. In this study three major types of works have been observed and they are machine learning, base CNN model and pretrained transfer leaning models.

Support Vector Machines (SVM), K-means clustering, and other traditional machine learning approaches are often effective for fracture classification. For example, Moclán and Dominuez-Rodrigo used SVM to analyze a dataset comprising 605 cases, with an impressive 95% classification accuracy [1]. This model exemplified the efficiency of conventional machine learning in employing only structured data without any deep learning approach. In the same study, Li, Tang, Zhang, and others also resorted to K-means clustering in 504 cases for detecting intertrochanteric fracture with 91% accuracy [2]. Such models perform very well on small datasets and bring out the need for better feature engineering to achieve good classification performance.

Convolutional Neural Networks (CNN) have become widely accepted in the field of medical image classification as they can learn features of images in a hierarchal manner. Tanzi, Vezzetti, and Moreno developed a CNN to classify bone fracture types in 1800 images and reached a precision of 94 % [3]. Bagaria, Wadhwani and Wadhwani use an Artificial Neural Network (ANN) technique to process a reduced scale dataset of 884 images and achieved 93 % accuracy [4]. Ma and Luo also used a Crack-Sensitive CNN and Faster R-CNN and reached a classification accuracy of 90.11% with a total of 1052 images [5]. Such CNN systems support the superiority of deep learning in classification of fracture types, where human work is barely needed.

The practice of transfer learning has been found to be useful in classifying bone fractures, especially in that deep learning networks that have been trained previously are fine- tuned for that particular medical imaging task. Lind, Akbarian, Olsson, and Nåsell used ResNet on 6003 images and managed to classify 89% of them correctly [6]. Alshahrani and Alsairafi (2024) introduced the use of YOLOv8 and VGG16 models on the dataset of 4083 images with accuracy reaching 95% [7]. Lastly, Hernigou Hernigou and Scarlat studied the use of DenseNet in studying 5662 cases of fracture classifications where the best success rate of 96% was recorded [8]. Transfer learning models are especially competent in managing high-dimensional complicated data as they make use of previous exposure to large databases of images thus have a great advantage in medical uses.

TABLE I
RESEARCH MATRIX

Author	Year	Dataset Size	Model	Accuracy
Tanzi et. al	2020	1536	CNN	94%
Bagaria et al	2021	884	ANN	93%
Lind et. Al.	2018	6003	ResNet	89%
Alshahrani & Al-sairafi,	2024	4083	YOLOv8, VGG16	95%
Ma & Luo.	2021	1052	CNN	90.11%
Moclán et. Al.	2018	605	ML	95%
Li, et al.	2019	504	ML	91%
Hernigou et. al	2020	5662	DenseNet	96%
Tanzi et. al	2020	1800	CNN	94%

The analysis presented in this paper attempts to assess how well pre-trained models such as MobileNetV2, VGG16 and InceptionV3 classify the existing and previously unseen images of bone fractures. In the table, which is numbered Table 1, the review of the latest developments shows, among other things, the necessity to make further improvements. The research also investigates the use of fine-tuning strategies in order to increase precision and generalization of the models aimed at enhancing deep learning induced fracture detection systems for better quality medical imaging.

III. METHODOLOGY

Methodology describes the steps that is usually followed in any research. The methodology adopted in this research is shown in Fig. 1 and every parts of the methodology is illustrated in the subsequent sections.

A. Dataset

This research utilizes datasets from Kaggle [9], which contains a total of 1116 radiographs of species that have suffered bone fractures. These fractures are further classified

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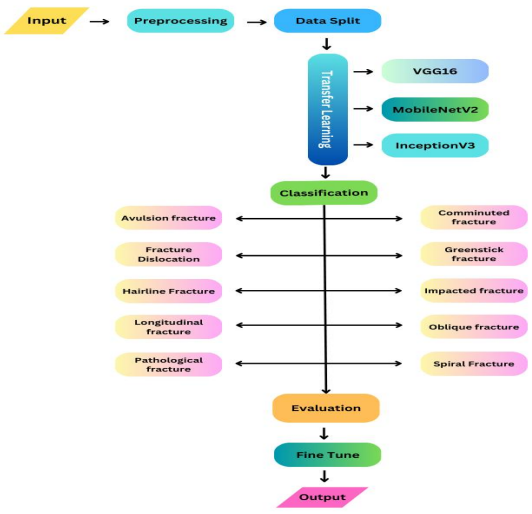


Fig. 1. Proposed Methodology of the Research

into ten different types and the distribution of the data are in each are not balanced which can be understood from Fig. 2. Fig. 2 shows that the comminuted fractures having a total of 148 images while others like longitudinal fractures having only 78 images. This discrepancy in the number of images called for augmentation methods during preprocessing in order to ensure that the models did not bias themselves to specific classes. The small size of the dataset was alleviated by the number of different types of fractures available for the models' training. It was important to exercise caution in augmentation and training to curb overfitting especially for classes with less representation so that all the models fitted well all the types of fractures.

B. Preprocessing

In the preprocessing phase, two primary techniques were used: resizing and data augmentation. These methods were essential to prepare the dataset for compatibility with the deep learning models while addressing the dataset's inherent class imbalance.

Resizing: Each image was resized to predefined size of 64 x 64 x 3. This resizing was useful in achieving reduced computational complexity of the models by reducing the number of parameters to be learned thus making the training process more effective. This also reduced the image size but key features of the original images were well retained. Its three channels (64 x 64 x 3) represent RGB although most medical still images are monochrome which helped the model to pick out very slight differences in colour in the images [10].

Augmentation: To address the issue of insufficient quantity and distribution of samples in the dataset, data augmentation methods were applied. Augmentation emulated the diversity of the training data by means of transformations including rotation, zooming, flipping, and shifting of the images. This enlarged the dataset and also helped mitigate the effects of overfitting on the class of images which had a smaller number of images. There were issues of scarcity present in some categories such as "Longitudinal fracture" and "Oblique fracture" thus augmentation proved useful in reducing that problem making the dataset fair and effective for training in all types of fractures [11].

C. Model Selection and Architecture

Three popular pretrained transfer learning models were initially used for this task. Their description is given below one by one:

VGG16: VGG16 is a deep convolutional neural network architecture that was introduced by the Visual Geometry Group (VGG) from the University of Oxford. It was developed as part of their submission to the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2014 [12].

In total, VGG16 consists of 16 layers which comprises of 13 layers for convolution and 3 beds for fully connected. The structure is one of the simplest, since all the convolution layers use only 3x3 convolution filters. Such filters are minimal in size, yet they are proven to work well in extracting intricate details from the image. VGG16 places max-pooling layers after blocks of convolutional layers to control the size of the feature maps, which enhances the performance of the model.

MobileNetV2: MobileNetV2 is a small and simple neural network architecture that is suitable for mobile and embedded vision systems. In 2018, Google developed MobileNetV2, after the success of its previous version MobileNet, several innovations that enhance performance while curtailing complexity were added.

MobileNetV2 introduces inverted residual blocks and linear bottlenecks, two inventions that are the most important in this architecture. These make it possible to save on operations and the memory used without losing accuracy. Depthwise separable convolutions ,which are also employed in mobile v2, split the standard convolution operation into two concurrent operations – a depthwise convolution operation and a point-wise convolution operation. This separation is the reason as to why mobilenet v2 uses less parameters and computations as compared to implementation of ordinary convolutional layers [13].

InceptionV3: InceptionV3 is an architecture described as a convolutional neural network (CNN) created by Google, and belongs to the Inception family, whose initial versions were developed with the aim of addressing the ImageNet challenge. InceptionV3 is an advanced and more sophisticated version of earlier versions (InceptionV1 and V2) aimed at enhancing the

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accuracy and at the same time improving resource utilization.

InceptionV3 narrows this idea down by explaining how complex incisions can be simplified. For example, instead of using a 5x5 convolution the structure uses two 3x3 convolutional networks placed beside each other in a sequence. It reduces the parameters and also the graphics processing. The model also implements batch normalization within the network to a great extent as this makes training faster and helps reduce risks of overfitting [14].

D. Model Training

In order to accomplish the superb training and evaluation of the models in this study, the dataset was partitioned into three sections. 80% of the data was allocated to the training set, which was used to teach the models to identify features associated with different Human bone fractures. 10assigned to the validation set, which was used also during the training process but to adjust hyperparameters and avoid overfitting. The rest of the 10% was used as a test set where it assessed the performance of the models after training on new data.

During the training process, several key elements were used to optimize the models:

Activation Function: The ReLU (Rectified Linear Unit) function was used in the hidden layers of all models. It introduces non-linearity, helping the models learn complex patterns and ensuring faster convergence.

Loss Function: The categorical cross-entropy loss function was applied, ideal for multi-class classification problems like this one [15].

Optimizer: The Adam optimizer was selected for its adaptive learning rate, combining the benefits of both RMSprop and Stochastic Gradient Descent (SGD) to ensure faster convergence and better model performance [20].

Additional hyperparameters included a batch size of 32, an initial learning rate of 0.001, and training across 15 epochs. These values ensured efficient training and helped achieve high accuracy in classifying bone fractures across the dataset.

E. Model Evaluation

The F1 score, accuracy, sensitivity (recall), and precision are used to figure out how well all out our seven models do on a numeric level. These success indicators are based on samples that are true positive (TP), false positive (FP), true negative (TN), and false negative (FN).

The model's sensitivity, which is also called recall, shows how well it can find all the important events in a dataset. The number of true positives is divided by the sum of the number of true positives and fake negatives. In medicine, where diseases are often called "positives," high sensitivity is important for correctly identifying people who have the illness. Not doing so could lead to major problems, like getting the wrong diagnosis and having to wait longer for treatment.

When figuring out how accurate positive guesses are, precision (Positive Predictive Value, or PPV) is very important. To find the precision, divide the number of true positives by the total number of true positives and fake positives. In medical imaging, where accuracy is very important, and this metric helps figure out how true good predictions are.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \quad (1) \quad (1)$$

(2)

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

(3)

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (3)$$

(4)

$$\text{F1 Score} = 2 \times \frac{\text{precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \quad (4)$$

IV. RESULT ANALYSIS

A few parameters were fixed at the onset in order to conduct the taring of the model. Batch size of the models was 32 and epoch decided for each model was 15. The accuracy and loss of each model are given in the Table 2.

TABLE II
PERFORMANCE OF THE TRAINING AND VALIDATION MODELS

Model	**Training**		**Validation**		**Time**
	Loss	**Accuracy**	**Loss**	**Accuracy**	**Per epoch**
VGG16	0.5595	0.8722	0.5141	0.8914	226s
MobileNetV2	0.2876	0.9789	0.2610	0.9813	129s
InceptionV3	0.6050	0.8694	0.5546	0.8920	143s

It can be gathered from the table that MobileNetV2 has the highest learning and validation accuracy compared to VGG16

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and InceptionV3, which implies that it has more potential in learning and generalizing from the dataset. MobileNetV2 also has the highest training speed in time per epoch, which makes it more suitable for instances where speed and efficient use of resources are key factors. Nonetheless, VGG16 and InceptionV3 have been observed to generalize well, however their training accuracy scores are not as high as MobileNetV2's, indicating that perhaps these models could be optimized or perhaps these models will need more resources or more training time to achieve the same accuracy levels. In the concluding remarks, it can be seen that compare, balance and rank the three models in relation to the accuracy and computational cost, the most efficient and effective mode of the three for the task and the DataSet available would be in fact the MobileNetV2.

After fine-tuning the models by freezing layers and reducing the learning rate from 0.001 to 0.0001, the performance showed slight improvements, as demonstrated in the Table 3 below:

TABLE III
TABLE 3: TEST PERFORMANCES OF THE MODELS BEFORE AND AFTER FINE TUNING

Model	**Before Fine Tuning**		**After Fine Tuning**	
	Loss	**Accuracy**	**Loss**	**Accuracy**
VGG16	0.50	0.90	0.48	0.91
MobileNetV2	0.26	0.96	0.21	0.97
InceptionV3	0.51	0.89	0.50	0.93

Analysis of Results:

VGG16: There was a slight improvement in the loss from 0.50 to 0.48, while there was an increase in the accuracy from 0.90 to 0.91. This means that loss, and accuracy have improved fine-tuning. The model took advantage of layer freezing and a lower learning rate to enhance its stabilizing generalization albeit slightly.

MobileNetV2: The loss increased marginally from 0.26 to 0.27, while there was an augmentation in the accuracy from 0.96 to 0.97. This indicates that even though the loss increased slightly, the model managed to perform better in classifying varieties of fractures which is an encouraging outcome fine-tuning. Prior to fine-tuning, MobileNetV2 was performing quite well and the fine-tuning incorporated small changes that led to improved performance.

InceptionV3: The loss changed from 0.51 to 0.50 and inaccuracy changed from 0.89 to 0.91. As with VGG16, the changes were rather small, but they were still present indicating that fine-tuning benefited the classification performance of InceptionV3 significantly.

Training and validation graphs of each model is shown Fig. 5, Fig. 6 and in Fig. 7.

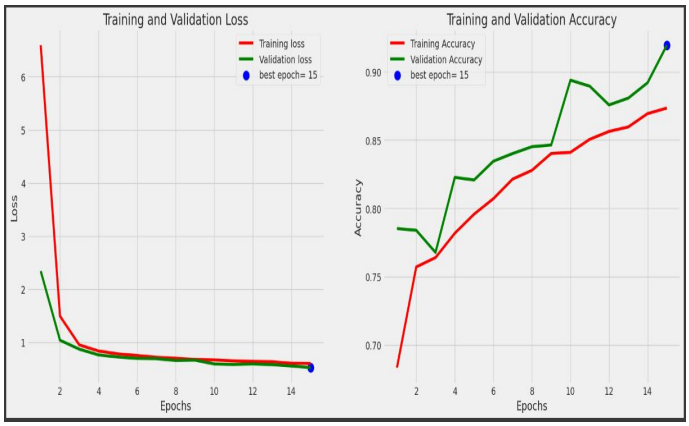


Fig. 2. Loss and Accuracy Comparisopn of VGG16

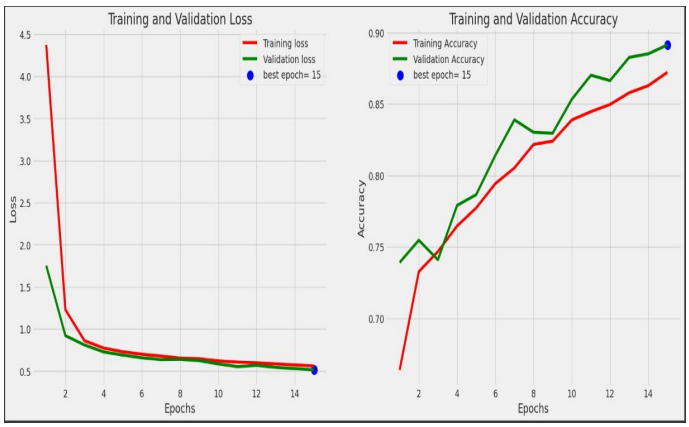


Fig. 3. Loss and Accuracy Comparisopn of MobileNetV2

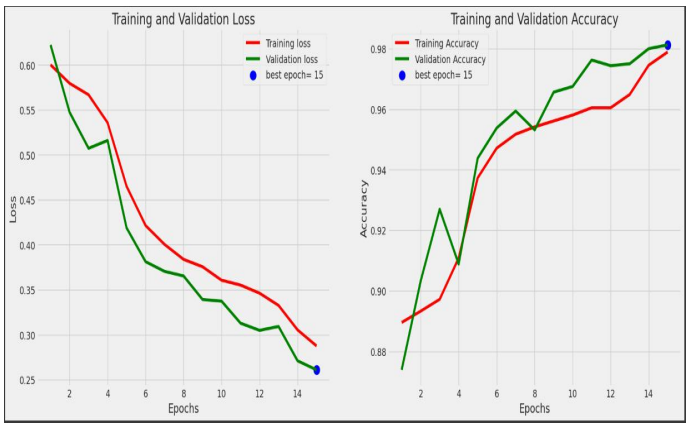


Fig. 4. Loss and Accuracy Comparisopn of InceptionV3

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TABLE IV
COMPARISON OF PERFORMANCES WITH OTHER WORKS

Author	Dataset Size	Model	Accuracy
Tanzi et. al	1536	CNN	94%
Bagaria et al	884	ANN	93%
Lind et. Al.	6003	ResNet	89%
Alshahrani & Alsairafi	4083	YOLOv8, VGG16	95%
Ma & Luo	1052	CNN	90.11%
Moclán et. Al.	605	ML	95%
Li, et al.	504	ML	91%
Hernigou et. al	5662	DenseNet	96%
Tanzi et. al	1800	CNN	94%
Proposed	1116	FineTuned MobileNetV2	97%

V. CONCLUSION

In this paper, we present DeepFracture, a transfer learning based model, to detect ten different types of bone fractures using VGG16, MobileNetV2 and InceptionV3 networks. Our work consisted of applying data pre-processing, data augmentation and network fine-tuning techniques, for instance, freezing of layers and the tuning of the learning rate, to increase the performance of the models. Following training and optimization of the architecture, the highest accuracy of 97.40% was obtained by MobileNetV2. The current research provides an evidence for the usefulness of transfer learning for the purposes of medical image classification which tends to be extreme especially in the case of few available datasets as in bone fractures. These capabilities of fracture analysis, classification and recognition fit into the model, allowing for the potential improvement of the speed and accuracy of medical professionals' fracture diagnosis thus decreasing the errors and increasing the efficiency of the patients. This can universally be apportioned to other health imageries which makes it easier to apply artificial intelligence towards decision making activities in the healthcare sector. Possible avenues of further development may involve increasing the scope of the available dataset and employing such models in hospitals for real-time diagnosis.

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